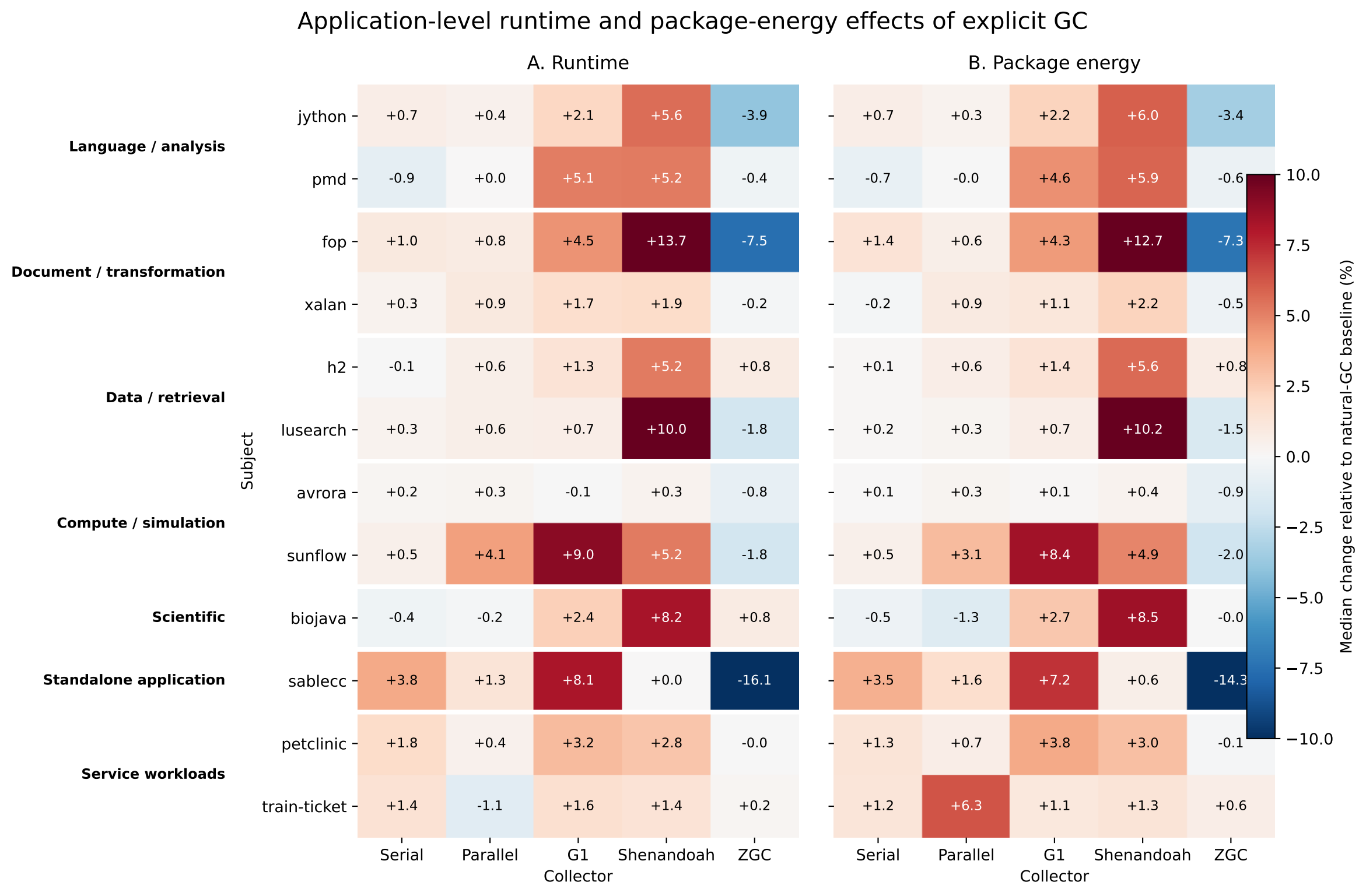
Application-level runtime and package-energy effects of explicit GC
A. Runtime
B. Package energy
Language / analysis
Document / transformation
Data / retrieval
Compute / simulation
Scientific
Standalone application
Service workloads
jython
pmd
fop
xalan
h2
lusearch
avrora
sunflow
biojava
sablecc
petclinic
train-ticket
Subject
+0.7 +0.4 +2.1 +5.6 -3.9
-0.9 +0.0 +5.1 +5.2 -0.4
+1.0 +0.8 +4.5 +13.7 -7.5
+0.3 +0.9 +1.7 +1.9 -0.2
-0.1 +0.6 +1.3 +5.2 +0.8
+0.3 +0.6 +0.7 +10.0 -1.8
+0.2 +0.3 -0.1 +0.3 -0.8
+0.5 +4.1 +9.0 +5.2 -1.8
-0.4 -0.2 +2.4 +8.2 +0.8
+3.8 +1.3 +8.1 +0.0 -16.1
+1.8 +0.4 +3.2 +2.8 -0.0
+1.4 -1.1 +1.6 +1.4 +0.2
+0.7 +0.3 +2.2 +6.0 -3.4
-0.7 -0.0 +4.6 +5.9 -0.6
+1.4 +0.6 +4.3 +12.7 -7.3
-0.2 +0.9 +1.1 +2.2 -0.5
+0.1 +0.6 +1.4 +5.6 +0.8
+0.2 +0.3 +0.7 +10.2 -1.5
+0.1 +0.3 +0.1 +0.4 -0.9
+0.5 +3.1 +8.4 +4.9 -2.0
-0.5 -1.3 +2.7 +8.5 -0.0
+3.5 +1.6 +7.2 +0.6 -14.3
+1.3 +0.7 +3.8 +3.0 -0.1
+1.2 +6.3 +1.1 +1.3 +0.6
Serial
Parallel
G1
Shenandoah
ZGC
Collector
Median change relative to natural-GC baseline (%)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0